Predictive Analysis & Real-Time Tracking of Carbon Emissions from Domestic Electrical Loads Using Gradient Boosting Model

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Abstract

Urban household appliances play a significant role in the rapid rise of CO_2 in the atmosphere which is a dominant gas causing global warming. However, there is a lack of tools for residents to monitor and reduce their impact in real time. To address this problem, this paper presents a novel framework trained on a comprehensive dataset of appliance level energy utilization of various households along with their metadata and the weather patterns of their neighbourhood. The system is trained using a Light Gradient Boosting Machine (LightGBM) model that accurately forecasts the CO_2 emissions in real time and provides user specific recommendations. The novel contribution of this work is the integration of the AI driven predictive methodology with a dynamic recommendation system having a conceptual framework for a solid sorbent carbon capture unit, which creates a complete monitoring and mitigate solution. The trained model demonstrated the ability to accurately track real world household emissions with a low Mean Absolute Error. An interactive dashboard was implemented utilizing the trained model as its core which provides users with dynamic data driven recommendations, tools for real time simulation and historical analysis of their carbon footprint. The proposed integrated system provides a user-friendly solution contributing to a big step towards urban decarbonization.

Keywords: Carbon Emissions Forecasting, Gradient Boosting, Carbon Capture, Predictive Engine, Carbon Footprint

1 Introduction and Literature Survey

The rapid rise of atmospheric carbon dioxide (CO2) is a major concern for the rapid rise of global warming. Global warming repercussions include major changes in temperature, precipitation, or wind patterns, among other effects, that occur over several decades or longer. Urban residential buildings play a critical role in this problem. According to Intergovernmental Panel on Climate Change (IPCC), urban settlements contribute to climate change, generating about 70% of global CO2 emissions [1]. To mitigate this problem, we need to start working at the root level, individual households. Traditional solutions to this problem such as monthly electricity bills not only provide a delayed and highly aggregated feedback but are also insufficient to make actionable changes in time. While modern smart home devices do offer some level of real time monitoring of appliances individually, they still lack real time monitoring or mitigation recommendations. This paper presents a novel solution which integrates a framework powered by Artificial Intelligence(AI) to monitor real time household CO2 emissions with a dynamic recommendation system for users to reduce or mitigate emissions. This paper is organized as follows. Section I reviews the literature in the field of household energy and CO2 emissions forecasting. Section II discusses about the worldwide CO2 emissions with respect to household appliances. Section III presents the methodology for data aggregation and model training. Section IV details the experimental results and model's performance. Finally, section V discusses the recommended methods and techniques, concluding with the future scope of this research.

1.1 Household Energy Monitoring and Disaggregation

Understanding the consumption of individual appliances is a major challenge in the analysis of household energy patterns. Recently, the field of Non-Intrusive Load Monitoring (NILM) is aiming to solve this problem by computationally disaggregating the total energy consumption of a household into individual appliance consumption [2]. Traditional methods such as event based methods have been replaced by modern methods like deep learning approaches such as Convolutional Neural Networks to recognize electrical usage profile of various household appliances [3]. While these focus on the disaggregation, our work utilizes a pre-disaggregated dataset bypassing the disaggregation problem overall for the novel purpose of training a direct, real time CO2 emissions forecasting and recommendations model, which is not typically addressed in the NILM method.

1.2 Machine Learning for Energy Forecasting

There are studies that employs a range of models such as traditional methods like Autoregressive Integrated Moving Average (ARIMA) [4] to more sophisticated algorithms such as Long Short-Term Memory (LSTM) networks and Gradient Boosting Machines [5]. The foundation of these models relies on features like historical energy patterns, timestamps, weather data to make predictions and the proposed system builds upon this foundation with a key distinction being the integration of household metadata such as occupancy, building type and size. This distinct approach allows our model to create a more context aware and scalable predictive framework.

1.3 Recommender Systems for Sustainable Choice

The study of sustainable behaviour centres around the concept of "eco-feedback" where users are provided with information about their energy consumption to influence their actions [6]. Effective systems present timely suggestions to make more sustainable choices which is significant improvement over delayed feedback relying on monthly electrical bills [7]. The proposed model applies these principles moving beyond simple historical reporting by coupling a real time predictive model with a dynamic recommender engine to provide an immediate and context specific feedback making it much more actionable than those of the traditional systems.

1.4 Direct Air Capture and Solid Sorbents Technologies

The concept of Direct Air Capture (DAC) represents an emerging field of mitigation technologies that work on directly removing of CO2 from the ambient air itself. An effective approach for this method is using solid sorbent materials like Zeolites or Metal Organic Frameworks which exhibit strong affinity towards adsorbing CO2 molecules [8].

2 Worldwide Co_2 Emissions from Household Appliances

The global effort to fight climate change begins with understanding the emission sources across different sectors of the society. The residential sector plays a major role in contributing to global energy consumption as well as the CO2 emissions associated with it.

2.1 Global Energy Landscape and the Role of the Residential Sector

Nearly 40% of global carbon dioxide emissions come from the real estate sector. Out of these emissions, approximately 70% are produced by building operations, while the remaining 30% comes from construction [9]. Within the real estate sector, residential buildings are the primary driver of this demand. As the population is increasing, the demand for energy load in households is also increasing for reasons such as cooling, heating, lighting and powering appliances which in turn make the decarbonization of the residential sector a critical priority to achieve the goal of Net Zero Emissions (NZE).

2.2 Breakdown of Household Energy End-Use

Space and water heating hold the largest share of energy in residential buildings globally which are followed by energy required to power several appliances and electronic devices [10]. However, it varies significantly based on climate, building quality, economic status as well as occupant behaviour which leads to a diverse residential load profile [11]. For example, in tropical areas, air conditioning and refrigeration are dominant loads whereas in cold regions, space heating account for more than half of a residential energy use. The connectivity of smart homes to manage both new and existing energy loads more intelligently and ensuring that technological advancements to a net reduction carbon footprint is the challenge right now.

3 Sources of Data & Methodology

3.1 Dataset Description

The base of the proposed system is the REFIT Electrical Load Measurements dataset, which contains high-frequency power data from 20 households in United Kingdom [12]. Data from nine individual appliance monitors (IAMs) at an 8-second resolution and an aggregate power reading are included for each home. Its features includes size, type of house, year of construction, and occupancy. To obtain secondary data for local weather conditions, data was selected as a geographically close proxy from a historical UK weather dataset. The features that were extracted for this study were the maximum temperature (°C), humidity (%), and rainfall (mm). The UK Department for Energy Security and Net Zero's official emission factor for the UK power grid, 0.233 $kgCO_2e/kWh$, was used to translate the estimated energy usage into a carbon footprint [13].

3.2 Data Aggregation and Preprocessing

A comprehensive dataset is created for model training using the REFIT and weather dataset. The dataset is characterized by the following:

- Total Data Points: The final, resampled dataset consists of 917,756 time-series entries.
- Total Features (X Labels): The model was trained using a total of 56 input features for all unique individual appliances monitored across the 20 households, such as 'Blender', 'Bread-maker', 'Chest_Freezer', 'Computer', 'Dehumidifier', 'Desktop_Computer', and household metadata like House_ID, Occupancy, and one-hot encoded features derived from building Type, Size, and Construction_Year and even weather data: max_temp °c, humidity %, and rain mm and also time-based features: Hour_of_Day, Day_of_Week, and Is_Weekend.
- Target Variable (y Label): The single target variable for the regression model is Aggregate_CO_2_kg, representing the total predicted CO_2 emissions in kilograms for each 15-minute interval.

3.3 Feature Engineering and Target Variable Formulation

To reduce computational cost and smooth out high-frequency noise, the entire dataset was resampled to a 15-minute frequency. Numerical columns were aggregated using their mean value for each interval, while categorical columns used the first observed value.

- Features Based on Time: Hour_of_Day, Day_of_Week, and Is_Weekend were created to document temporal patterns.
- Categorical Features: Text-based metadata features (Type, Size, and Construction_Year) were one-hot encoded into a numerical representation using the pandas get_dummies function. The Occupancy and House_ID fields were converted to integer types.

A two-step formula was then used to calculate the target variable, Aggregate_CO2_kg. Initially, energy in kWh was calculated by converting the average power in Watts for every 15-minute interval (0.25 hours):

$$\text{Energy (kWh)} = \frac{\text{Average Power (W)} \times 0.25 \text{ (h)}}{1000} \tag{1}$$

Second, the final objective was obtained by multiplying this energy value by the grid emission factor for the UK i.e. 0.233:

$$CO_2 \text{ Emissions (kg)} = \text{Energy (kWh)} \times \text{Emission Factor (kg}CO_2\text{e/kWh)} \quad (2)$$

3.4 Predictive Modelling

A high-performance gradient boosting framework called Light Gradient Boosting Machine (LightGBM) was chosen as the proposed predictive model. This model was selected because of its demonstrated accuracy on tabular data, computational speed, and efficiency in handling the large, feature-rich dataset created in the earlier phases.

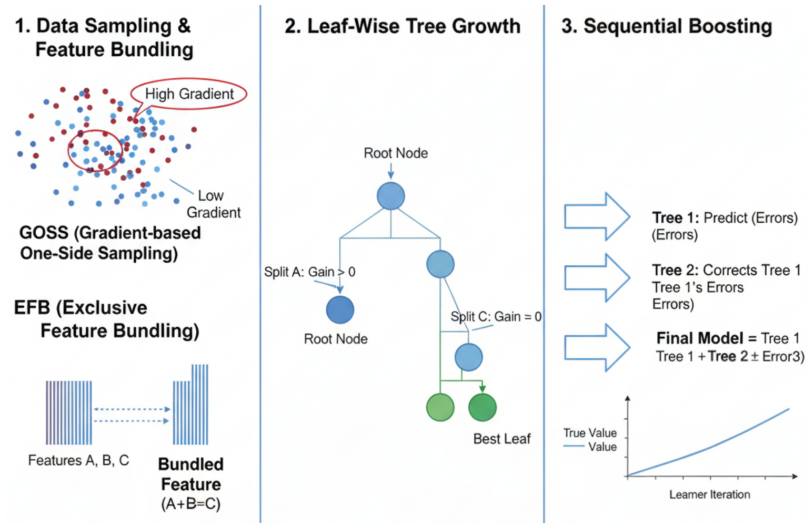

Fig. 1 Conceptual Overview of the LightGBM model

LightGBM operates on the principle of boosting, where multiple weak prediction models (decision trees) are sequentially trained as seen in 1. The major difference of LightGBM from other gradient boosting frameworks like XGBoost is its innovative approach to tree construction:

3.5 Model Building and Prediction

The dataset was divided chronologically to guarantee a reliable assessment that replicates a real-world forecasting situation. The first 80% of the time-series data was used to create the training set, and the remaining 20% was used to create the test set. Three performance metrics: R-squared (R2), Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) were used to evaluate the model's performance.

4 Results and Model Performance

4.1 Feature Importance Analysis

To understand the key factors of emissions learned by the model, a feature importance analysis was conducted which can be seen in Figure 2 which displays the top 15

most influential features, ranked by their contribution to the model's predictions. The microwave and kettle are examples of high-power, short-duration appliances that rank highly among individual appliances, indicating that their impact on emission peaks is disproportionate.

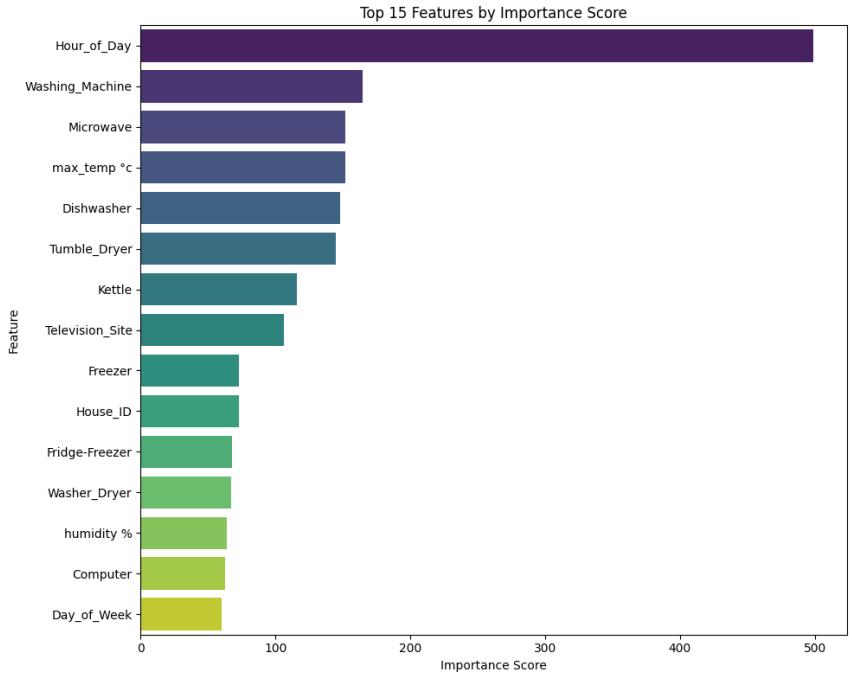

Fig. 2 Top 15 Features by Importance Score from the LightGBM Model

4.2 Quantitative Performance Evaluation

To objectively assess the predictive capability of the LightGBM model, three standard regression evaluation metrics were employed: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-squared (R^2).

$$\mathbf{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \quad (3) \qquad \mathbf{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \quad (4) \qquad \mathbf{R^2} = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \quad (5)$$

MAE measures the average magnitude of errors. **RMSE** penalizes larger errors more heavily due to the squaring of differences, making it sensitive to significant deviations. **R-squared** indicates how well the model explains the variability in the target. The model was evaluated on a test set, comprising the 20% of the data in chronological order. The results, as shown in Table 1, tells us about the model's strong prediction capability. The low MAE of 0.0131 kg indicates that, on average, the model's prediction for a 15-minute interval is off by only 13.1 grams of CO . The model is

more severely penalized for a few larger errors, which most likely correspond to the underestimation of sharp emission peaks, even though most predictions are extremely accurate, as indicated by the RMSE, which is roughly double the MAE.

Table 1 Model Performance Metrics

Metric	Value
MAE ($kgCO_2$)	0.0131
RMSE ($kgCO_2$)	0.0260
R-squared (R^2)	0.4376

4.3 Visual Analysis of Predictive Performance

Beyond aggregated metrics, analysing the distribution of errors (residuals) provides crucial insights into a regression model's performance.

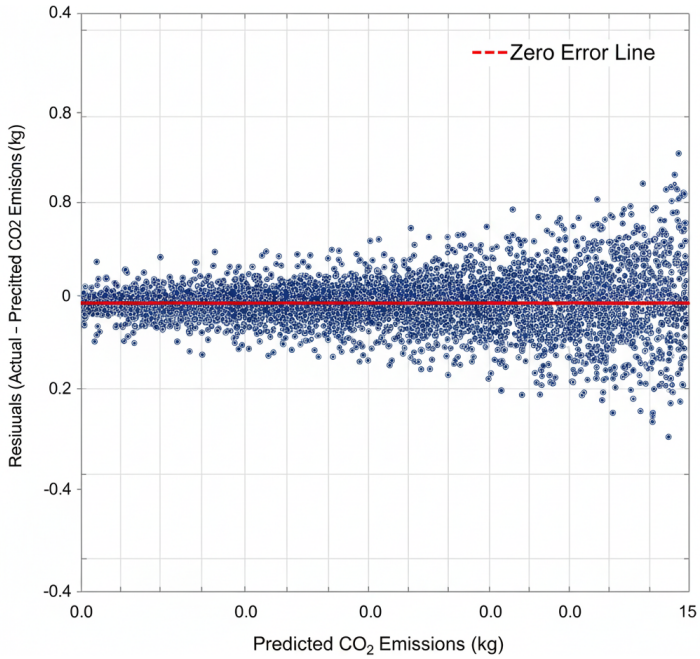

Fig. 3 Comparison of Actual vs. Predicted CO Emissions using Residual Plot

A residuals plot visualizes the difference between the actual and predicted values against the predicted values (or sometimes against a feature). Figure 3 shows the distribution of residuals (actual - predicted CO emissions) against the predicted CO emissions. A random scatter around zero suggests a robust model with no systematic biases.

Further, Figure 4 illustrates the predicted emissions which were plotted against the actual emissions for a sample one-week period from the test set to qualitatively

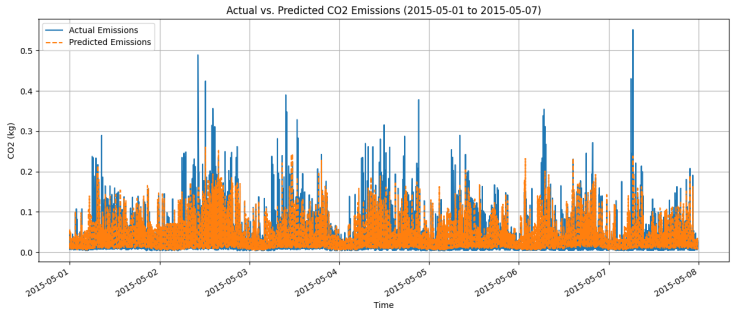

Fig. 4 Comparison of Actual vs. Predicted CO Emissions over a Sample Week in the Test Set

evaluate the model's performance. The plot provided visual confirmation of the earlier quantitative results. The projected emissions (dashed line) accurately reflect the typical consumption patterns of the household by closely following the overall trend and diurnal rhythm of the actual data (solid line). Nevertheless, it is also clear that the model tends to underestimate the size of the strongest, highest emission spikes.

5 Recommended Methods and Techniques for Household Emission Reduction

The suggested pre-established and emerging methods for reducing household energy consumption and the associated CO emissions are broadly categorized into behavioural interventions, technological efficiency upgrades, and advanced smart home strategies. Real-time digital eco-feedback effectively promotes energy savings by influencing occupant behaviour. The effectiveness of "nudging" systems depends on using high-quality, timely, and context-aware information to subtly guide users toward sustainable choices without restricting their options [14]. Moreover, significant energy reductions can be achieved through technological upgrades, such as replacing traditional lighting with energy-efficient LEDs, upgrading to modern high-efficiency appliances, and adopting electric heat pumps over conventional heating systems [15]. Smart technologies optimize home energy via load automation [16] (e.g., smart thermostats adjusting HVAC based on occupancy [17]) and participation in Demand Response (DR) programs [18], all enabled by granular data from devices like smart plugs [19].

6 Conclusion and Future Scope

In this work, a novel, integrated AI-powered framework is developed to monitor and mitigate household CO emissions. By aggregating a complex dataset from 20 distinct households with appliance-level readings and household metadata and integrating the local weather data, a robust predictive model is built. The model trained using LightGBM presented high performance, accurately forecasting short-term emissions in real-time and predicting the patterns of residential energy consumption.The model's predictive power presented user-based, interactive dashboard which offers historical analysis and real-time simulation with dynamic data-driven recommendations. Future

work will focus on advancing the system through both hardware integration and algorithmic improvements. A primary next step involves developing a physical carbon capture unit managed by a microcontroller to create an automated, closed-loop system. This will be followed by real-world deployment in test homes, utilizing live data and formal user studies to measure the system's true impact on energy savings and occupant behaviour.

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